

Q1.

David, a zoologist, is investigating a particular species of monitor lizard. He measures the lengths, in centimetres, of a random sample of this particular species of lizard. His measured lengths are

53.2 57.8 55.3 58.9 59.0 60.2 61.8 62.3 65.4 66.5

The lengths may be assumed to be normally distributed.

David correctly constructed a 90% confidence interval for the mean length of lizard using

$$\bar{x} \pm \left(b \times \frac{s}{\sqrt{n}} \right)$$

the measured lengths given and the formula

This interval had limits of 57.63 and 62.45, correct to two decimal places.

- (a) State the value for b used in David's formula.

(1)

- (b) David interprets his interval and states,

"My confidence interval indicates that exactly 90% of the population of lizard lengths for this particular species lies between 57.63 cm and 62.45 cm".

Do you think David's statement is true? Explain your reasoning.

(2)

- (c) David's assistant, Amina, correctly constructs a $\beta\%$ confidence interval from David's random sample of measured lengths.

Amina informs David that the width of her confidence interval is 8.54.

Find the value of β .

(3)

(Total 6 marks)

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Q2.

Dimitra is an athlete who competes in 400 m races. The times, in seconds, for her first six races of the 2012 season were

54.86 53.09 53.75 52.88 51.97 51.81

- (a) Assuming that these data form a random sample from a normal distribution, construct a 95% confidence interval for the mean time of Dimitra's races in the 2012 season, giving the limits to two decimal places.

(5)

- (b) For the 2011 season, Dimitra's mean time for her races was 53.41 seconds. After her first six races of the 2012 season, her coach claimed that the data showed that she would be more successful in races during the 2012 season than during the 2011 season. Make **two** comments about the coach's claim.

(2)

(Total 7 marks)

Q3.

Gemma, a biologist, studies guillemots, which are a species of seabird. She has found that the weight of an adult guillemot may be modelled by a normal distribution with mean μ grams. During 2012, she measured the weight, x grams, of each of a random sample of 9 adult guillemots and obtained the following results.

$$\sum x = 8532 \quad \text{and} \quad \sum (x - \bar{x})^2 = 38\,538$$

- (a) Construct a 98% confidence interval for μ based on these data.

(5)

- (b) The corresponding confidence interval for μ obtained by Gemma based on a random sample of 9 adult guillemots measured during 2011 was (927, 1063), correct to the nearest gram.

- (i) Find the mean weight of guillemots in this sample.

(1)

- (ii) Studies of some other species of seabird have suggested that their mean weights were less in 2012 than in 2011. Comment on whether Gemma's two confidence intervals provide evidence that the mean weight of guillemots was less in 2012 than in 2011.

(2)

(Total 8 marks)

Q4.

At the start of the 2012 season, the ages of the members of the Warwickshire Acorns Cricket Club could be modelled by a normal random variable, X years, with mean μ and standard deviation σ .

The ages, x years, of a random sample of 15 such members are summarised below.

$$\sum x = 546 \quad \text{and} \quad \sum (x - \bar{x})^2 = 1407.6$$

- (a) Construct a 98% confidence interval for μ , giving the limits to one decimal place.

(6)

- (b) At the start of the 2005 season, the mean age of the members was 40.0 years.

Use your confidence interval constructed in part (a) to indicate, with a reason, whether the mean age had changed.

(2)

(Total 8 marks)

Q5.

A factory produces bottles of brown sauce and bottles of tomato sauce.

- (a) The content, Y grams, of a bottle of brown sauce is normally distributed with mean μ_Y and variance 4.

A quality control inspection found that the mean content, y grams, of a random

sample of 16 bottles of brown sauce was 450.

Construct a 95% confidence interval for μ_Y .

(3)

- (b) The content, X grams, of a bottle of tomato sauce is normally distributed with mean μ_X and variance σ^2 .

A quality control inspection found that the content, x grams, of a random sample of 9 bottles of tomato sauce was summarised by

$$\sum x = 4950 \quad \text{and} \quad \sum (x - \bar{x})^2 = 334$$

- (i) Construct a 90% confidence interval for μ_X .

(5)

- (ii) Holly, the supervisor at the factory, claims that the mean content of a bottle of tomato sauce is 545 grams.

Comment, with a justification, on Holly's claim. State the level of significance on which your conclusion is based.

(3)

(Total 11 marks)

Q6.

Jim, a mathematics teacher, knows that the marks, X , achieved by his students can be modelled by a normal distribution with unknown mean μ and unknown variance σ^2 .

Jim selects 12 students at random and from their marks he calculates that $\bar{x} = 64.8$ and $s^2 = 93.0$.

- (a) (i) An estimate for the standard error of the sample mean is d . Show that $d^2 = 7.75$.

(2)

- (ii) Construct an 80% confidence interval for μ .

(3)

- (b) (i) Write down a confidence interval for μ , based on Jim's sample of 12 students, which has a width of 10 marks.

(1)

- (ii) Determine the percentage confidence level for the interval found in part (b)(i).

(4)

(Total 10 marks)

Q7.

The error, X °C, made in measuring a patient's temperature at a local doctors' surgery may be modelled by a normal distribution with mean μ and standard deviation σ .

The errors, x °C, made in measuring the temperature of each of a random sample of 10 patients are summarised below.

$$\sum x = 0.35 \quad \text{and} \quad \sum (x - \bar{x})^2 = 0.12705$$

Construct a 99% confidence interval for μ , giving the limits to three decimal places.

(Total 5 marks)

Q8.

Jane, who supplies fruit to a jam manufacturer, knows that the weight of fruit in boxes that she sends to the manufacturer can be modelled by a normal distribution with unknown mean, μ grams, and unknown standard deviation, σ grams.

Jane selects a random sample of 16 boxes and, using the t -distribution, calculates correctly that a 98% confidence interval for μ is (70.65, 80.35).

- (a) (i) Show that the sample mean is 75.5 grams. (1)
- (ii) Find the width of the confidence interval. (1)
- (iii) Calculate an estimate of the standard error of the mean. (3)
- (iv) Hence, or otherwise, show that an unbiased estimate of σ^2 is 55.6, correct to three significant figures. (2)
- (b) Jane decides that the width of the 98% confidence interval is too large.
Construct a 95% confidence interval for μ , based on her sample of 16 boxes. (2)
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- (c) Jane is informed that the manufacturer would prefer the confidence interval to have a width of at most 5 grams.
- (i) Write down a confidence interval for μ , again based on Jane's sample of 16 boxes, which has a width of 5 grams. (1)
- (ii) Determine the percentage confidence level for your interval in part (c)(i). (3)
- (Total 13 marks)

Q9.

Bishen believes that the mean weight of boxes of black peppercorns is 45 grams. Abi, thinking that this is not the case, weighs, in grams, a random sample of 8 boxes of black peppercorns, with the following results.

44 44 43 46 42 40 43 46

- (a) (i) Construct a 95% confidence interval for the mean weight of boxes of black peppercorns, stating any assumption that you make. (6)
- (ii) Comment on Bishen's belief. (2)
- (b) (i) Abi claims that the mean weight of boxes of black peppercorns is less than 45 grams. Test this claim at the 5% level of significance. (6)
- (ii) If Bishen's belief **is** true, state, with a reason, what type of error, if any, may have occurred when conclusions to the test in part (b)(i) were drawn. (2)
- (Total 16 marks)**

Q10.

A speed camera was used to measure the speed, V mph, of John's serves during a tennis singles championship.

For 10 randomly selected serves,

$$\sum v = 1179 \quad \text{and} \quad \sum (v - \bar{v})^2 = 1014.9$$

where $\bar{v}$ is the sample mean.

- (a) Construct a 99% confidence interval for the mean speed of John's serves at this tennis championship, stating any assumption that you make. (7)
- (b) Hence comment on John's claim that, at this championship, he consistently served at speeds in excess of 130 mph. (1)
- (Total 8 marks)**

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Q11.

The weight of fat in a digestive biscuit is known to be normally distributed.

Pat conducted an experiment in which she measured the weight of fat, x grams, in each of a random sample of 10 digestive biscuits, with the following results:

$$\sum x = 31.9 \quad \text{and} \quad \sum (x - \bar{x})^2 = 1.849$$

- (a) (i) Construct a 99% confidence interval for the mean weight of fat in digestive biscuits. (5)
- (ii) Comment on a claim that the mean weight of fat in digestive biscuits is 3.5 grams. (2)

- (b) If 200 such 99% confidence intervals were constructed, how many would you expect **not** to contain the population mean?

(1)

(Total 8 marks)

Q12.

The marks achieved by Pat in her homework assignments may be assumed to be normally distributed with mean μ .

The marks achieved by Pat in a random sample of 8 assignments were recorded as follows:

60 65 62 67 69 71 63 66

Construct a 99% confidence interval for μ .

(Total 5 marks)

Q13.

Alan's journey time, in minutes, to travel home from work each day is known to be normally distributed with mean μ .

Alan records his journey time, in minutes, on a random sample of 8 days as being

36 38 39 40 50 35 36 42

Construct a 95% confidence interval for μ .

(Total 5 marks)

Q14.

Ten students each independently carried out the same experiment in order to measure, in m s^{-2} , the value of g , the acceleration due to gravity, with the following results:

9.75 9.72 9.71 9.69 9.66 9.70 9.72 9.71 9.69 9.65

- (a) Assuming that values from the experiment are normally distributed, with mean g , construct a 95% confidence interval for g .

(6)

- (b) It was subsequently discovered that the equipment used in the experiment was faulty. As a consequence, each of the values above is 0.10 m s^{-2} less than the actual value.

Use this additional information to write down a revised 95% confidence interval for g .

(2)

(Total 8 marks)

Q15.

Members of a residents' association are concerned about the speeds of cars travelling through their village. They decide to record the speed, in mph, of each of a random

sample of 10 cars travelling through their village, with the following results:

33 27 34 30 48 35 34 33 43 39

- (a) Construct a 99% confidence interval for μ , the mean speed of cars travelling through the village, stating any assumption that you make.

(7)

- (b) Comment on the claim that a 30 mph speed limit is being adhered to by most motorists.

(3)

(Total 10 marks)

Q16.

The weight, W grams, of rice contained in a packet can be modelled by a normal distribution with mean μ and variance σ^2 .

The weights, in grams, of rice contained in each of a random sample of 10 packets were

301.2	305.6	298.4	297.7	292.6
306.1	296.8	299.2	304.0	308.4

- (a) Calculate unbiased estimates for μ and σ^2 .

(2)

- (b) Hence construct a 99% confidence interval for μ , giving the limits to one decimal place.

(5)

- (c) The packets are claimed to contain, on average, at least 290 grams of rice.

Comment on this claim.

(2)

(Total 9 marks)

Q17.

The time, T minutes, that parents have to wait before seeing a mathematics teacher at a school parents' evening can be modelled by a normal distribution with mean μ and standard deviation σ .

At a recent parents' evening, a random sample of 9 parents was asked to record the times that they waited before seeing a mathematics teacher.

The times, in minutes, are

5 12 10 8 7 6 9 7 8

- (a) Construct a 90% confidence interval for μ .

(7)

- (b) Comment on the headteacher's claim that the mean time that parents have to wait

before seeing a mathematics teacher is 5 minutes.

(2)

(Total 9 marks)

Q18.

The weights of lions kept in captivity at Wildcat Safari Park are normally distributed. The weights, in kilograms, of a random sample of five lions were recorded as

46 48 57 49 54

- (a) Construct a 95% confidence interval for the mean weight of lions kept in captivity at Wildcat Safari Park.

(6)

- (b) State the probability that this confidence interval does **not** contain the mean weight of lions kept in captivity at Wildcat Safari Park.

(1)

(Total 7 marks)

Q19.

The heights, in metres, of a random sample of 10 students attending Higrade School are recorded below.

1.76 1.59 1.54 1.62 1.49 1.52 1.56 1.47 1.75 1.50

Assume that the heights of students attending Higrade School are normally distributed.

- (a) Calculate unbiased estimates for the mean and variance of the heights of students attending Higrade School.

(3)

- (b) Construct a 90% confidence interval for the mean height of students attending Higrade School.

(5)

(Total 8 marks)

Q20.

The weights of oranges sold by a greengrocer are known to be normally distributed. The greengrocer takes a random sample of 10 oranges and records their weights in grams. The results are as follows.

176 194 193 191 170 177 178 160 188 169

- (a) (i) Calculate unbiased estimates of the mean and variance of the weights of oranges, giving your answers to one decimal place.

(3)

- (ii) Construct a 90% confidence interval for the mean weight of oranges.

(5)

- (b) The greengrocer currently sells the oranges by weight at a price of £1.20 **per kilogram**.

- (i) Deduce a 90% confidence interval for the mean price per orange. (2)
- (ii) The greengrocer decides to sell the oranges individually at a price of 20 pence per orange. Use the confidence interval you found in part (b)(i) to comment on this decision. (2)
- (Total 12 marks)

Q21.

Charles is an athlete specialising in throwing the javelin. During practice, he throws the following distances, in metres.

40.3	39.8	41.6	42.8	39.0	38.6	40.8	41.1
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- (a) Calculate unbiased estimates of the mean and the variance of the distance thrown. (3)
- (b) (i) Hence calculate a 95% confidence interval for the mean distance thrown. (5)
- (ii) State **two** assumptions that you need to make in order to do this calculation. (2)
- (Total 10 marks)

Q22.

A medical statistician is studying the weights, x kg, of newborn babies in a hospital. She finds that, in a certain month, 25 babies were born in this hospital and that for these babies, $\Sigma x = 88.2$ and $\Sigma x^2 = 326.2$.

- (a) (i) Calculate unbiased estimates of the mean and variance of the weights of babies born in this hospital. (4)
- (ii) Assuming that weights are normally distributed, calculate a 99% confidence interval for the mean weight of babies born in this hospital. (4)
- (b) She subsequently reads in a report that the mean weight of babies born in her region is 4.1 kg. Comment on her results with reference to this report. (2)
- (Total 10 marks)

Q23.

The contents, in grams, of a random sample of eight jars of yeast extract from a large batch were as follows:

342 347 346 343 344 347 344 345

The contents of the jars in the batch can be assumed to be normally distributed. Calculate a 90% confidence interval for the mean contents of the jars in the batch. Give your answer

to an appropriate number of significant figures.

(Total 8 marks)

Q24.

The dissolved oxygen contents, in parts per million, of samples of water taken from a river were as follows:

4.9 3.6 4.3 4.2 3.7 5.0 3.5 3.9 4.4

The data may be regarded as a random sample from a normal distribution.

Calculate a 95% confidence interval for the mean dissolved oxygen content of water in the river.

(Total 7 marks)

Q25.

A company manufactures components for the motor industry. The components are designed to have a length of 135.0 mm. Each day a technician takes a random sample of components and calculates a 95% confidence interval for the mean length.

- (a) On a particular Monday the technician takes a sample of nine components. Their lengths, in millimetres, are as follows.

135.1 135.7 134.9 135.2 136.3 135.7 135.9 136.0 135.6

Assuming the lengths of components produced on this Monday may be modelled by a normal distribution, calculate a 95% confidence interval for the mean length. Give your answer to an appropriate degree of accuracy.

(8)

- (b) On a particular Friday a random sample of 60 components had a mean length of 135.8 mm and a standard deviation of 3.9 mm. Calculate a 95% confidence interval for the mean length of components produced on this Friday.

(4)

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- (c) The company overhauls all machines when the confidence interval calculated does not contain 135.0 mm. State whether increasing the sample size, n , is likely to increase, decrease or leave unaltered:

- (i) the width of the confidence interval;

(1)

- (ii) the frequency of machine overhauls, explaining your answer.

(2)

- (d) Using your results in parts (a) and (b) as an example, or otherwise, explain why it is possible that production on a day when the confidence interval does not contain 135.0 mm will sometimes be more satisfactory than production on a day when the confidence interval does contain 135.0 mm.

(2)

(Total 17 marks)

Q26.

A health food co-operative imports a large quantity of dates and packs them into plastic bags labelled 500 grams. George, a Consumer Protection Officer, checked a random sample of the bags and found the weights, in grams, of the contents were as follows.

497 501 486 502 492 508 489 494

- (a) Assuming that weights follow a normal distribution, calculate, for the mean weight of contents of all the bags:
- (i) a 95% confidence interval; (8)
 - (ii) an 80% confidence interval. (2)
- (b) George is uncertain what conclusion to draw and so decides to start again by taking a much larger sample of the bags. He takes a random sample of 95 bags and finds they have a mean weight of 499.6 grams and a standard deviation of 9.3 grams.
- Use this larger sample to calculate a 90% confidence interval for the mean weight of contents of all the bags. (4)
- (c) The health food co-operative also imports raisins. George intends to take a random sample of 500 gram packets of raisins, weigh the contents and use the results to calculate an 80% and a 95% confidence interval for the mean weight, μ , of the contents of all the co-operative's packets of raisins.
- (i) Find the probability that:
 - (A) the 80% confidence interval contains μ . (1)
 - (B) the 95% confidence interval contains μ but the 80% confidence interval does not. (2)
 - (ii) Instead of calculating both confidence intervals from the same sample, George now decides to calculate the 95% confidence interval from one sample and the 80% confidence interval from a second independent random sample. Find the probability that the 95% confidence interval contains μ but the 80% confidence interval does not. (2)
- (Total 19 marks)**

Q27.

Applicants to join a police force are tested for physical fitness. Based on their performance, a physical fitness score is calculated for each applicant. Assume that the distribution of scores is normal.

- (a) The scores for a random sample of ten applicants were

55 23 44 69 22 45 54 72 34 66

Calculate a 99% confidence interval for the mean score of all applicants.

(8)

- (b) The scores of a further random sample of 110 applicants had a mean of 49.5 and a standard deviation of 16.5.

Use the data from this second sample to calculate:

- (i) a 95% confidence interval for the mean score of all applicants;
- (ii) an interval within which the score of approximately 95% of applicants will lie.
- (c) By interpreting your results in parts (b)(i) and (b)(ii), comment on the ability of the applicants to achieve a score of 25.
- (d) Give **two** reasons why a confidence interval based on a sample of size 110 would be preferable to one based on a sample of size 10.
- (e) It is suggested that a much better estimate of the mean physical fitness of all recruits could be made by combining the two samples before calculating a confidence interval. Comment on this suggestion.

(4)

(2)

(3)

(2)

(3)

(Total 22 marks)

Q28.

A firm is considering providing an unlimited supply of free bottled water for employees to drink during working hours. To estimate how much bottled water is likely to be consumed, a pilot study is undertaken. On a particular day-shift, ten employees are provided with unlimited bottled water. The amount each one consumes is monitored. The amounts, in ml, consumed by these ten employees are as follows:

110 0 640 790 1120 0 0 2010 830 770

- (a) Assuming the data may be regarded as a random sample from a normal distribution, calculate a 95% confidence interval for the mean amount consumed on a day-shift.
- (b) (i) Give a reason, based on the data collected, why the normal distribution may not provide a suitable model for the amount of free bottled water which would be consumed by employees of the firm.
- (ii) A normal distribution may provide an adequate model but cannot provide an exact model for the amount of bottled water consumed. Explain this statement giving a reason which does not depend on the data collected.
- (c) Following the pilot study the firm offers free bottled water to all the 135 employees who work on the night-shift. The amounts they consume on the first night have a mean of 960 ml with a standard deviation of 240 ml.
- (i) Assuming that these data may be regarded as a random sample, calculate a

(8)

(3)

90% confidence interval for the mean amount consumed on a night-shift.

(4)

- (ii) Give **one** reason why it may be unrealistic to regard the data as a random sample of the amounts that would be consumed by all employees if the scheme was introduced on all shifts on a permanent basis.

(1)

(Total 16 marks)



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